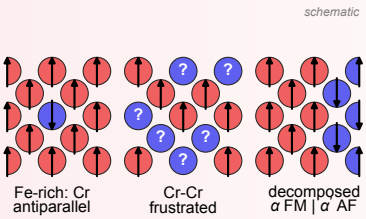
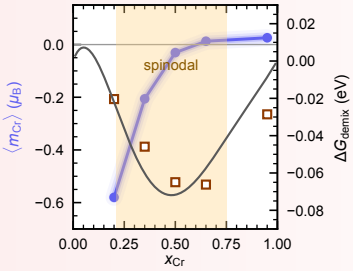
FRUSTRATE

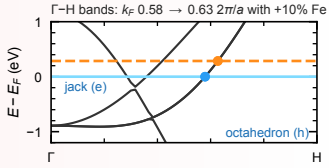
(a) atoms & moments

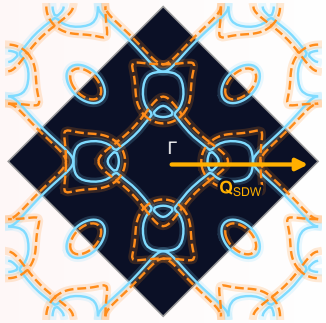




NEST

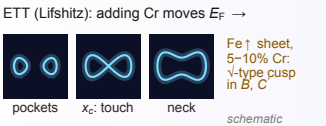
(b) Cr Fermi surface vs Fe content

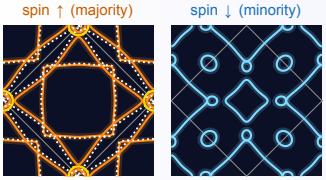




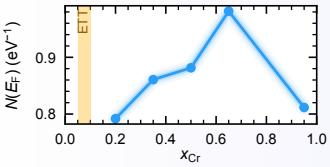
TOPOLOGY

(c) Fe-rich ETT: spin ↑ vs ↓



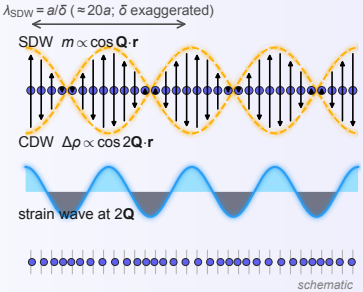


Cr shifts the majority states (band filling alone moves E_F away) ⇒ ETT at 5–10% Cr in spin ↑ only (Korzhavyi 2009); unfolded SQS Fermi surfaces pending

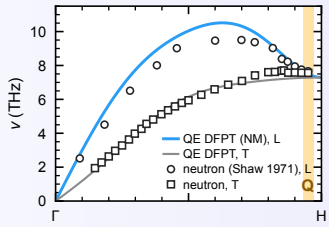


MODULATE

(d) SDW/CDW soften the lattice



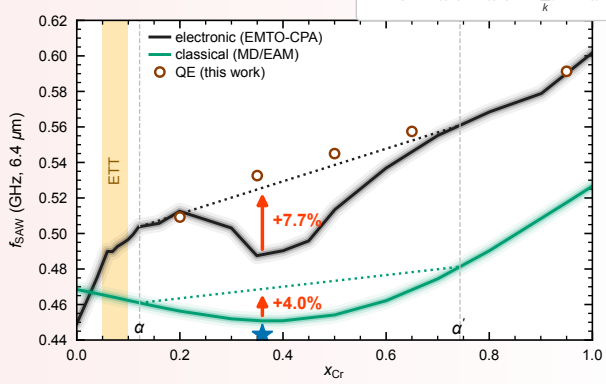
Kohn softening at **Q**:
lattice imprint of the SDW/CDW



STIFFEN

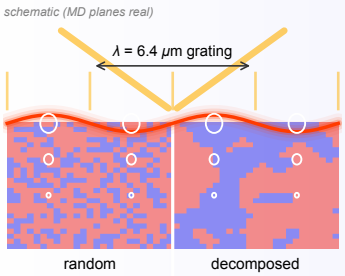
(e) electron → lattice: decomposition

$$C = C_{\text{ion}} + C_{\text{band}},\quad C_{\text{band}} \propto -\sum_k (\partial\varepsilon_k/\partial\eta)^2\,\delta(\varepsilon_k - E_F)$$



PROPAGATE

(f) Rayleigh wave in TGS



$C_{ij}\uparrow \Rightarrow G\uparrow \Rightarrow v_R = \eta(\nu)\sqrt{G/\rho}\uparrow \Rightarrow f = v_R/\lambda\uparrow$

Fe–36Cr, MD: 451 → 474 MHz (+5.1%)

electronic lever rule: +7.7%

DETECT

(g) (P)SAW vs pristine CASS

